

SECTION A – CASE QUESTIONS (Total: 50 marks)

Answer **ALL** of the following questions. Marks will be awarded for logical argumentation/ calculation and appropriate presentation of the answers.

CASE

Flying Success Ltd ("FSL"), a subsidiary of Dragon Group Ltd ("DGL"), is a company listed on the Hong Kong Stock Exchange that specialises in organising and conducting drone demonstration shows and providing drone rental services to the Government of the Hong Kong SAR and various business organisations. Due to the slowdown in the economy, the Government and businesses in Hong Kong have been searching for new business opportunities. The low-altitude economy ("LAE") appears to be emerging as a potential driver to stimulate economic growth, and the Hong Kong Government has recently formed different working groups to develop and regulate drone business opportunities. One of the key LAE areas that the Government has significantly supported in Hong Kong and the Greater Bay Area is drone delivery services, which is a service that FSL would like to invest in.

Drone delivery service involves upstream activities like scientific research and drone manufacturing, midstream activities such as component supply, drone repair, and maintenance, and downstream activities, including drone shows, tourism, and new delivery services. As such, LAE involves new aviation infrastructure, driving systems, intelligent connectivity technology, automation, and artificial intelligence, which have all been developing rapidly over the past few years. Yet the lack of pre-existing low altitude airway networks and routes, air traffic control, and drone parking has made it challenging to develop LAE for highly populated cities like Hong Kong.

The public has been increasingly accepting of drone services thanks to drone demonstration shows and the use of drones in rescue activities, and any members of the public have also been learning to operate drones at leisure. As such, drone education has been provided to increase public awareness and understanding of this emerging economy, particularly regarding safety and security concerns, such as drone accidents, privacy breaches, and personal data security breach issues. LAE has been praised for its low carbon emission and low air pollution, both of which could help lower the city's carbon footprint. Nevertheless, the operation of drones creates noise pollution, the disposal of drones and batteries creates environmental pollution, and there have been reports of hill fires having been caused by drone malfunctions.

LAE is an emerging economy with several regulatory uncertainties concerning privacy and data security, product liability, and civil aviation concerns. The operation of drones has created significant liabilities to the public that require prevention and mitigation. Given that LAE is an emerging economy, FSL needs to invest significant amounts of resources to plan and develop its drone delivery service business while also being aware of the uncertainty of returns.

FSL has been organising drone demonstration shows for several years, and the business is growing very quickly due to the Hong Kong Government's initiatives to further the mega event economy to attract tourists. Due to the rapid growth of the business, FSL has not implemented any performance measurement systems to measure performance during drone demonstration shows, though FSL has sufficient money to operate their drone demonstration shows with limited amounts of overdraft facilities. FSL is also concerned about the sustainability of the drone demonstration show business due to strengthened rules and laws governing LAE. FSL wants to set up a multidimensional performance measurement system to use key performance indicators to assess their business performance so that they can improve their drone demonstration show business.

FSL has a minimal amount of inventory, and all sales and purchases are on credit. FSL has observed an increase in new customers requesting drone demonstration shows, but they have few returning customers. There have also been several bottlenecks in operations that have previously led to cancellations of drone demonstration shows at the last minute. Recent failures of the global positioning system due to climate factors have also led to the cancellation of several drone demonstration shows, requiring FSL to offer compensation to their customers, who are now increasingly doubting if they should use FSL's services again. Moreover, the attractiveness of FSL's drone demonstration shows has declined due to repetitiveness and overlapping content.

FSL is preparing a proposal to apply for the Innovation and Technology Fund ("ITF") under the Innovation and Technology Commission to further develop their drone demonstration show business and to plan for a future drone delivery service business. The financial controller of FSL has asked the newly recruited finance manager to prepare the proposal within one week so that they can meet the application deadline. The finance manager, a certified public accountant, has been working in the finance department of a local bank for 10 years. He has not worked in emerging businesses before, so he has limited knowledge regarding LAE and drone service businesses.

The financial controller has also reminded the finance manager to prepare a profitable yet realistic financial forecast for the proposal, despite FSL currently operating at a breakeven financial position, to increase the chance of securing the required funding, which is one of the key criteria for the finance manager to successfully pass his job probation and continue working for FSL. The finance manager has a nephew that works at ITF and is planning to treat the nephew to a luxury dinner and buy him some gifts in exchange for advice on how to effectively prepare the proposal to successfully secure the funding.

FSL is considering a new financing structure to provide sufficient funding for the development of their drone delivery service business. As FSL is in an emerging economy and is in a growth phase, they would prefer to achieve a lower cost of equity to support their new business development.

FSL has two possible plans. Plan A involves obtaining a long-term bank loan that would account for 20% of their total capital structure, while Plan B involves obtaining a long-term bank loan that would account for 50% of their total capital structure. The industry asset beta of comparable companies is assumed to be 0.8, with a market risk premium of 5% and a risk-free rate of 3%. Additionally, FSL has proposed a financing structure that includes issuing ordinary shares amounting to HK\$5,000,000, with a required rate of return equivalent to the cost of equity, and obtaining a long-term bank loan of HK\$1,250,000 at a preferred interest rate of 10% per annum, supported by a corporate guarantee from DGL. Based on this financing structure, FSL estimates earnings per share ("EPS") of HK\$2 at the beginning of the new financing plan, with an estimated EPS growth rate of 20% in year 1 and 30% annually from year 2 onwards. The dividend payout ratio is assumed to remain consistent at 20% from year 2, with no dividends declared in year 1. The profits tax rate is 16.5%.

Question 1 (15 marks – approximately 27 minutes)

Acting as the financial controller of FSL, write a memo to the Board of Directors to evaluate external influences, such as legal and environmental, among others, affecting the company's new drone delivery service business in the low-altitude economy in Hong Kong.

Note: A maximum of 2 marks for communication skills will be awarded.

(15 marks)

Question 2 (10 marks – approximately 18 minutes)

Acting as the financial controller of FSL, propose an appropriate multidimensional performance measurement system to the Board of Directors. Recommend and evaluate TWO specific performance indicators for each aspect in the performance measurement system to measure the performance of FSL's drone demonstration show business.

(10 marks)

Question 3 (10 marks – approximately 18 minutes)

Identify and explain the ethical issues faced by the finance manager in preparing the proposal to apply for the Innovation and Technology Fund and advise on the appropriate actions the finance manager should take to mitigate the issues.

(10 marks)

Question 4 (15 marks – approximately 27 minutes)

- (a) Calculate the cost of equity of Plan A and Plan B. Evaluate the results of both plans and advise the financial controller to choose a suitable plan for FSL.

Note: A maximum of 2 marks for analytical skills will be awarded.

(6 marks)

- (b) Assuming FSL adopts Plan B, calculate the share price of FSL under the new financing structure.

(9 marks)

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End of Section A

SECTION B – ESSAY / SHORT QUESTIONS (Total: 50 marks)

Answer **ALL** of the following questions. Marks will be awarded for logical argumentation/ calculation and appropriate presentation of the answers.

Question 5 (21 marks – approximately 38 minutes)

LLD is an airline company listed on the Hong Kong Stock Exchange and based in Hong Kong. It leverages the geographical advantage of Hong Kong as a major travel hub to operate routes within Asia. During COVID-19, many airlines faced financial difficulties due to very low valuation. LLD aimed to strengthen its footprint in Southeast Asia by leveraging its strong financial position to acquire SEAA, a Southeast Asia budget airline based in Malaysia. SEAA owns many aircrafts, which LLD believed would create good synergy to expand its capacity after the acquisition. LLD had conducted comprehensive due diligence, including legal and finance due diligence with reputable lawyers and accounting firms, and believed this acquisition would create significant synergies for LLD.

However, after 3 years, even with a significant rebound in international travel, the management of LLD have realised that the acquisition of SEAA has not delivered the synergy it expected, and the financial performance of SEAA has been worse than before COVID-19, which has resulted in SEAA making a loss and achieving negative cash flow over the last 2 years.

Required:

- (a) Advise on FIVE common causes of acquisition failures and explain how each could potentially lead to the failure of the acquisition of SEAA. (12 marks)
- (b) As SEAA is incurring losses and negative cash flow, creating financial pressure on LLD, LLD is considering selling SEAA. Evaluate various valuation models and identify appropriate methods and a suitable model for the sale. (9 marks)

Question 6 (16 marks – approximately 29 minutes)

LCT is a technology company based in Hong Kong with functional currency in US Dollars. Recently, LCT have focused on the development of artificial intelligence ("AI") applications and have set up a research centre in Mainland. The development of AI requires significant computational power and physical space to house equipment for high-density data centres.

LCT currently leases external data centres in Mainland for its storage needs. LCT expects to pay RMB15 million in the coming year to lease their data centres and expects rent to increase by 10% per year due to rising demand and cost inflation. Considering the price increase and the Chinese government's upcoming subsidies for AI development investments, LCT is considering building their own data centre with the support of the Chinese government to enhance its future growth.

The following information regarding building a data centre in Mainland is summarised below:

1. The Chinese government will provide LCT land and a building at no cost for four years from the start of the project.
2. LCT is required to invest RMB5.56 million for renovation of the data centre.
3. Hardware and equipment estimated at RMB37 million will be paid by LCT at the end of the first year.
4. It will take one year for the renovation, ordering, and installation of the hardware before the centre can fully operate.
5. Operating costs will be RMB7 million per year from the start of the centre's operation.
6. LCT's cost of capital is 7%.
7. LCT is expected to resell the data centre at RMB18 million at the end of the fourth year from the start of the project.

Required:

- (a) Calculate the net present value and the IRR of the data centre, and advise if LCT should accept the project. Assume cash flow happens at the end of each year.
(10 marks)

- (b) Calculate the percentage change in the renovation costs (RMB5.56 million) which would result in the opposite recommendation given in Question 6(a).
(3 marks)

LCT took on a loan of RMB50 million from a domestic bank in Mainland, which will be fully repaid at the end of the sixth year. LCT would like to hedge its foreign exchange exposure by entering into a 6-year forward contract for USD/RMB. The spot rate of USD to RMB is 7.2, and the interest rates of USD and RMB are 3% per annum and 1.5% per annum, respectively.

- (c) Considering the interest of the TWO currencies, calculate the theoretical 6-year forward contract rate of USD/RMB.
(3 marks)

Question 7 (13 marks – approximately 23 minutes)

Shadow, a logistics company incorporated in Hong Kong and listed on the mainboard of the Hong Kong Stock Exchange, provides domestic logistic services to various commercial customers. Shadow started its sustainability reporting last year. Shadow reported a 20% growth in orders received from customers last year. However, due to keen pricing competition, Shadow's margins have been under significant pressure, causing Shadow to be very cautious when purchasing new vehicles. By maximising the use of its existing vehicles, the efficiency of Shadow's fleet have declined.

Below is a summary of Shadow's vehicle usage.

	2025	2024
Number of trucks	41	38
Kilometres travelled per truck per year	18,000 km	16,000 km
Average kilometre per litre of diesel by truck	5 km per litre	7 km per litre
Carbon emission per litre of diesel fuel	2,600 kg/L	2,700 kg/L

Carbon credits currently trade at HK\$50 per ton (1 ton is equal to 907 kg).

Sustainability requires a significant commitment, including financial and human resources. Given Shadow's current business context, Shadow is reluctant to invest further; instead, Shadow plans to present its sustainability activities in a "smart way", via the following proposal:

1. Stop disclosing carbon emissions from trucks used to provide logistics service to customers. Alternatively, Shadow plans to disclose carbon emissions of electric vehicles used by its management only, as the carbon emissions from trucks are significantly higher than those of management's used electric vehicles.
2. Reduce the scale of tree planting activities from involving the entire company staff to including only Shadow's board of directors (i.e. seven people), allowing Shadow to save on labor.
3. Publish an "environmental certificate" on Shadow's website, without details such as certification process, accreditation framework, etc. The certificate will be issued by a friend of Shadow's board of directors who does not have any Environmental, Social and Governance ("ESG") certification experience.

Required:

- (a) Calculate Shadow's carbon emissions in 2025 and 2024 and advise on how much Shadow need to pay if Shadow would like to fully neutralise its trucks' carbon emissions by buying carbon credit.

(4 marks)

- (b) By referring to examples of greenwashing discussed in Vollero's 2022 book "Greenwashing: Foundations and Emerging Research on Corporate Sustainability and Deceptive Communication", justify whether the above proposal could be considered greenwashing.

(9 marks)

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